

Bio-test - 1Section - AQuestion - 1

(b) Sertoli Cells

Question - 2

Apomixis is a form of asexual fertilisation in which seeds are formed without physical fertilisation.

Question - 3

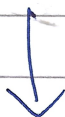
(a) Point mutation

Question - 4

Reverse transcriptase

Question - 5

(b) Random mating

Section - BQuestion - 6

Question-6Active Immunity

- Produced by the host's immune system when exposed to an antigen
- Slow in response
- Eg: -

Passive Immunity

- Produced when ready-made antibodies are directly introduced into the body.
- Fast in response
- Eg: -

Question-9

Two essential features of an ideal contraceptive:

- User-friendly: ~~and~~ Easy to use and understand
- Availability: It should be easily available everywhere.

Question-7

Primary treatment: Physical process involving the removal of small and large suspended particles through filtration & sedimentation

Secondary: Biological process using aeration tanks in the presence of flocs.

Section - CQuestion - 11

Segregation of the members of any pair of alleles is independent of the segregation of other pairs in the formation of reproductive cells. This is the law of ~~the~~ Mendel's Law of Independent Assortment.

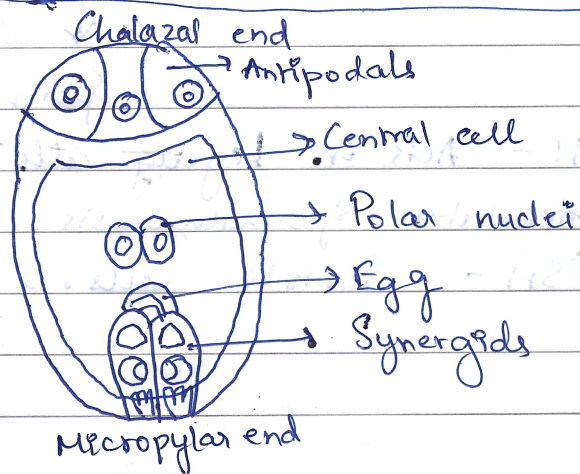
Phenotypic ratio: $9:3:3:1$

Question - 13

- (a) Mutualism: An interaction where both interacting species benefit mutually.
Example: Lichens (Algae & Fungi) - symbiotic relationship
- (b) Commensalism: Interaction in which 1 species benefits while the other is neither harmed or benefited.
Example: Barnacles growing on a whale
- (c) Parasitism: Interaction where 1 organism benefits but the other is harmed.
Example: Mosquito on a human

Question - 12

Mature Angiosperm
embryo sac



Section-D

(Option B)

(a)

- Multiplication phase - Spermatogonia ($2n=46$) present on the inside wall of seminiferous tubules multiply by mitotic division.
- Growth phase - Some spermatogonia enlarge to form primary spermatocytes ($2n$)
- Maturation phase - A Primary spermatocyte undergoes a meiotic division to form 2 equal haploid spermatids (i.e. secondary spermatocytes ($n=23$)). Each of these secondary spermatocytes undergo a secondary meiotic division to produce 4 equal spermatids (n).
- Spermiogenesis - The ~~spermatids~~ non-motile spermatids ~~enlarge~~ become spermatozoa (sperms)

Spermatogonia ($2n$)

↓ (Mitotic division)

Primary spermatocyte ($2n$)

↓ (Meiosis I)

Secondary spermatocyte (n)

↓ (Meiosis II)

Spermatids (n)

↓ (Spermiogenesis)

Spermatozoa (n)

- (b) LH - Acts on Leydig cells to stimulate testosterone, which stimulates spermatogenesis
- FSH - Sertoli cells ... ?